Introduction to Economics

Class 11 — Production and costs

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slide 1 of 26

Slide 1 Title

Block 4's first class. Classes 8 to 10 looked at whole markets: surplus and taxes, controls and trade, and the cases where a market gets it wrong. From today the camera moves inside one firm: what it costs to make things, how that cost changes as the firm makes more, and what changes once it can grow. One small firm, a print shop, carries the whole class; a real price that fell to zero opens it, and the day's tools answer it after the break.

Quiz 8 — Graphs

20 minutes

Two sheets, both sides: Part 1 is the first sheet, Part 2 the second. Answer both.

Write each number in its box and the labels inside their areas on the graph; calculations in the Working space.

Write your name on both sheets.

At the end: photograph the code at the top left (your result appears there), then hand in *both sheets*.

quiz page, not numbered

Quiz Start of class, on paper

A graph quiz on paper opened the class: two sheets, Part 1 on the first and Part 2 on the second, each question with its own grid and space for working. The numbers went into the printed boxes and the labels inside their areas on the graph; the code at the top of each page was photographed before both sheets were handed in, and the marked quiz appears at that link.

Plan for today

- What economists count as a cost: explicit and implicit costs, opportunity cost
- From workers to output: the production function
- Total, average and marginal cost
- Why the cost curves are U-shaped
- Short run and long run; economies and diseconomies of scale

Slide 2 Plan for today

The class at a glance. Before the break, one small firm from its bills to its cost per bag: what economists count as a cost, how output grows as workers are added to a single press, and the three measures of cost that follow from that. After the break, the pictures those numbers make, and what changes when the firm can change its size as well as its staff.

Two halves of fifty minutes with a ten-minute break; the opening question about electricity is answered in the second half.

Warm-up: what does the concert cost you?

Tonight: a concert or your supermarket shift?

Ticket paid: 30, no refund · shift pays 45 · bus to the concert: 2

1. Your opportunity cost of going?
   45 + 2 = 47 euros: the ticket is sunk
2. A friend offers 40 for the ticket. Now?
   going costs 87: the 40 is given up too
3. The concert is worth 60 to you. Go?
   no offer: go (60 > 47); with the offer: sell and work (60 < 87)

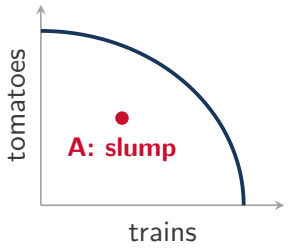
Slide 3 Warm-up: what does the concert cost you?

A warm-up on Class 2's opportunity cost, with a concert against a supermarket shift. The cost of going is what going makes you give up tonight, and the test for each number is whether it changes with the choice: the shift's pay and the bus fare do, while the ticket money is gone whatever you do, so it is sunk and no part of the answer. A thing you already own can still have an opportunity cost, namely what someone would pay you for it, so once a buyer appears, going means turning that offer down too. Whether to go then compares what the evening is worth to you with what it costs, and a larger cost can flip the answer.

Warm-up: what moves the frontier?

A country makes trains and tomatoes

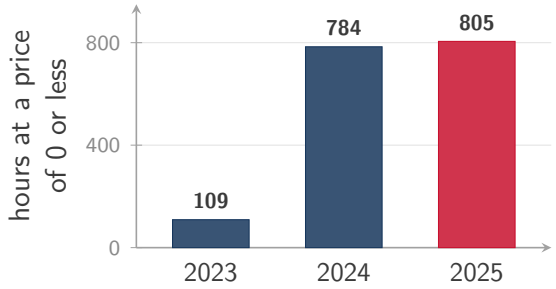
1. A greenhouse method doubles the tomatoes per worker
   the tomato end only: the frontier swings out
2. Workers arrive who can do either job
   both ends move out
3. Factories idle in a slump reopen
   a move toward the frontier, not a shift



Slide 4 Warm-up: what moves the frontier?

The second warm-up is Class 3's frontier, on trains and tomatoes. The frontier is the most of each good the country could make with everything in use, so it moves only when that changes, and a method that helps one good changes only that good's end of it. Workers who can do either job raise what is possible on both axes. A slump leaves the country inside the frontier, at the point marked on the figure; putting idle factories back to work changes where the country sits, not what it could make, and that is the trap the third item sets.

Spain, 2025: about 800 hours of electricity at a price of zero or less



Almost all in the middle of the day, most in spring.

Why would anyone sell electricity for nothing?

Own count from the day-ahead prices of OMIE (the Spanish–Portuguese wholesale electricity market), mainland Spain (set per quarter-hour since October 2025: four of them count as one hour).

Slide 5 Spain, 2025: about 800 hours of electricity at a price of zero or less

The hook, counted from the Iberian market operator's own price files. OMIE runs the market where Spain's and Portugal's generators sell the next day's electricity, hour by hour (quarter-hour by quarter-hour since October 2025), and in 2025 the price for mainland Spain was zero or below in about 805 of the year's hours, roughly one hour in eleven, almost all of them around midday and most of them in spring; in 2023 it had happened in 109. Your bill at home was not zero, because it adds the grid, taxes and charges; what fell to nothing was the wholesale price. The question is the slide's: why would a firm keep selling its product for nothing, or even pay to sell it? Keep it open; slide 14 answers it with the day's tools.

Counting what the owner gives up: the shop earns 1,700, not 4,400

The print shop's month

20 days · 4 printers · 220 bags a day at 4 euros

	euros
sales	17,600
− explicit costs (rent and press, printers)	13,200
= accounting profit	**4,400**
− implicit costs (her pay at an agency, interest forgone)	2,700
= economic profit	**1,700**

Accountants count the money paid out; economists count every cost given up.

Slide 6 Counting what the owner gives up: the shop earns 1,700, not 4,400

Meet the firm the whole class follows: a print shop with one press, printing customers' designs on bags they bring. The table counts its month twice. The accountant subtracts only money paid out; the economist also subtracts what the owner gives up by being here, the pay she could earn elsewhere and the interest her savings could earn invested elsewhere, which is why the reading says a firm's costs are its opportunity costs. The smaller profit is the one that says whether the shop is worth running: positive, so it beats her best alternative. The savings themselves are not a monthly cost; only the interest they could earn is.

In pairs, three minutes: a yoga studio's profit

A yoga studio's year

sales	72,000
rent, teachers, heating	40,000

The owner left a job paying 29,000. Her 75,000 of savings in the studio could earn 4 % a year.

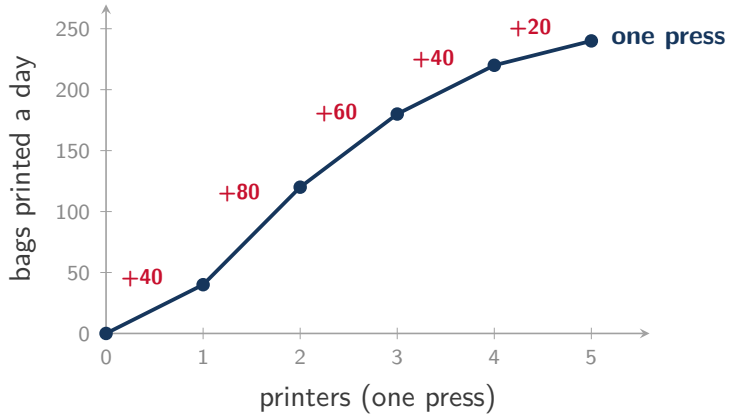
1. Accounting profit?
   72,000 − 40,000 = 32,000
2. Implicit costs?
   29,000 + 3,000 = 32,000
3. Economic profit?
   0: sales just cover every cost
4. Economic profit 0: is she losing money?
   no: she earns what her best alternative pays

Slide 7 In pairs, three minutes: a yoga studio's profit

Exercise 1 is slide 6 on another firm. The first profit uses only the two numbers in the table; the second question asks what the owner gives up that nobody invoices, her old salary and the return her savings could earn, and the savings themselves stay out of it because she still has them. Here the two sums happen to match, so the economic profit is zero, and the last item is the point: zero means the studio pays her exactly what her best alternative would, not that she is losing money. If that alternative improved, the same studio would start to lose by comparison, and closing would pay.

One press, more printers: from the third on, each adds fewer bags

A print shop in Poblenou: one press; customers bring the bags.

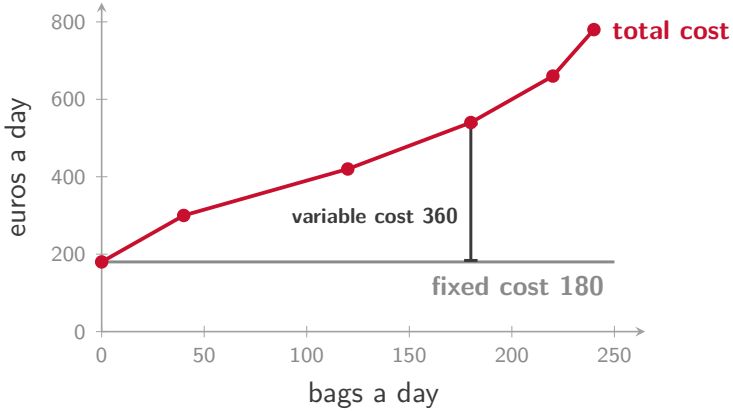


Marginal product: the bags one more printer adds. It falls once they crowd the press.

Slide 8 One press, more printers: from the third on, each adds fewer bags

The shop's production function: how many bags a day each number of printers can make with the press and the workshop fixed. What matters is not the total but the step between the points, the extra bags one more printer adds. It rises at first, because two people can split the work, and then falls, because every extra printer waits for the same press and gets in the others' way; that fall is what the reading calls diminishing marginal product, and it arrives in any firm that adds one input while another stays fixed. Total output never falls here; it is the added bags that shrink, which is the slip to avoid.

Total cost climbs faster as the press gets crowded



From the third printer on, each adds 120 but fewer bags: the curve gets steeper.

Slide 9 Total cost climbs faster as the press gets crowded

The same production function turned into money. Rent and the press cost the same whether the shop prints nothing or 240 bags, so their line is flat; each printer adds the same amount to cost, so total cost climbs by equal steps as printers are added, but from the third printer on each step buys fewer bags than the last, which is why the curve bends upward to the right. The gap between the two lines at any output is the variable cost, the printers' pay. A flattening production function and a steepening cost curve are one fact seen from two sides. Fixed does not mean fixed for ever: it means not changing with output; the landlord can still raise the rent next year.

Per bag: the average bottoms out at 3.00; once the press is crowded, extra bags get dearer

printers	bags a day	total cost TC	AFC FC ÷ bags	AVC VC ÷ bags	ATC TC ÷ bags	MC extra cost ÷ extra bags
0	0	180	—	—	—	
1	40	300	4.50	3.00	7.50	3.00
2	120	420	1.50	2.00	3.50	1.50
3	180	540	1.00	2.00	**3.00**	2.00
4	220	660	0.82	2.18	**3.00**	3.00
5	240	780	0.75	2.50	3.25	6.00

FC, VC: fixed, variable cost · AFC, AVC, ATC: average fixed, variable, total cost · MC: marginal cost

Fixed cost 180 a day · each printer 120 a day.

MC = 120 ÷ the extra bags: it rises as each printer adds fewer.

Slide 10 Per bag: the average bottoms out at 3.00; once the press is crowded, extra bags get dearer

The two questions a firm asks about cost, side by side: what the typical bag costs, and what one more costs. The average has two parts pulling opposite ways, the rent spread over ever more bags and the printers' pay per bag, which falls while teamwork helps and rises once the press is crowded; the sum falls, then climbs, and its lowest value marks what the reading calls the efficient scale. The last column is not total cost divided by bags but the rise in total cost divided by the extra bags it buys; since every printer costs the same, that is his pay divided by his marginal product, which is the link back to slide 8. Where the extra bags cost exactly the average, the average stands still, which is why its lowest value appears twice.

In pairs, four minutes: an ice-cream maker's costs

An ice-cream maker's day

tubs a day	total cost
0	60
10	100
20	130
30	180
40	260

1. Its fixed cost?
   60 euros a day: the cost of making nothing
2. ATC at 30 tubs?
   180 ÷ 30 = 6.00 euros a tub
3. MC of the tubs from 30 to 40?
   80 ÷ 10 = 8.00 euros a tub
4. The rent rises by 30 a day. ATC at 30? MC from 30 to 40?
   ATC 7.00; MC still 8.00

Slide 11 In pairs, four minutes: an ice-cream maker's costs

Exercise 2 is slides 9 and 10 on a table with five rows. The fixed cost is whatever the maker pays at zero output; the average at any row is that row's total over its tubs; the marginal cost is the rise between two rows over the tubs between them, so the step of ten must be divided out. The last item is the one to remember: a higher rent adds the same amount to every row, so it raises the average at every output, but the rise between two rows is untouched, so marginal cost does not change. A cost paid whatever the output cannot change the cost of one more unit, and the second half of the class builds on exactly that.

Three honey farms: how much does the market supply? (optional)

Jars a week each farm would sell

	at 6 euros	at 8 euros
Farm A	300	400
Farm B	0	250
Farm C	0	150

1. Market supply at 8 euros?
   800 jars: add the jars
2. At 6 euros?
   300: only A sells
3. A fourth farm sells 100 at either price. At 6?
   400 jars a week

Slide 12 Three honey farms: how much does the market supply? (optional)

An optional exercise on Class 5's market supply, tied to what firms decide to sell. A market's supply at a price is what all its sellers would bring at that same price, so the move is to pick one row of the table and add across the farms; the prices are never added, and a farm that would sell nothing at a price contributes nothing at it, whatever it would do at a higher one. A new seller simply adds its quantity to the total at each price.

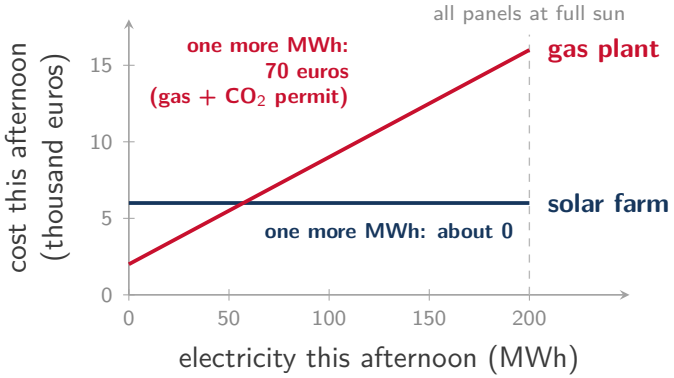
Break

Back at 11:10

Slide 13 Break

Ten minutes. After the break: the answer to the electricity question, then the print shop's costs as curves.

A solar farm's cost is almost all fixed: one more MWh costs about nothing



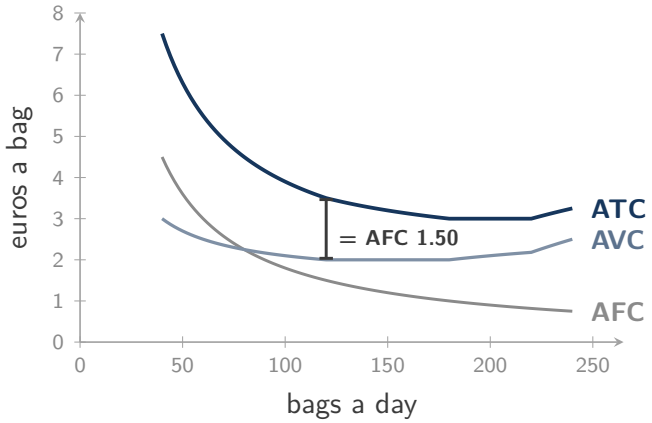
At a price of 0 the gas plant stops; switching off saves the solar farm nothing.

Illustrative numbers for one afternoon, not data.

Slide 14 A solar farm's cost is almost all fixed: one more MWh costs about nothing

Slide 5's answer, with slide 11's last lesson: a solar farm's costs are its panels, land, grid connection and the loan that paid for them, all owed whether or not it produces this afternoon, so its total cost is a flat line and one more megawatt-hour costs it next to nothing. A gas plant burns fuel for every unit and, since Class 10's permits, pays for the CO_2 as well, so its line climbs. At a price of zero the gas plant saves real money by stopping; the solar farm saves nothing, because its costs are already paid. On sunny spring middays there is enough near-free electricity to meet demand, which is why the price falls to zero; the farm pays for its panels in the hours when the price is above zero. So the reason is the cost structure, not that anyone pays the farm whatever the price. What this class does not explain is the "or less" of slide 5: a farm with these costs would rather switch off below zero, and the hours below zero come from sellers that a contract or a support scheme pays for every unit they produce, or that would lose more by stopping and restarting.

Fixed cost spread thinner, a crowded press: average total cost is a U

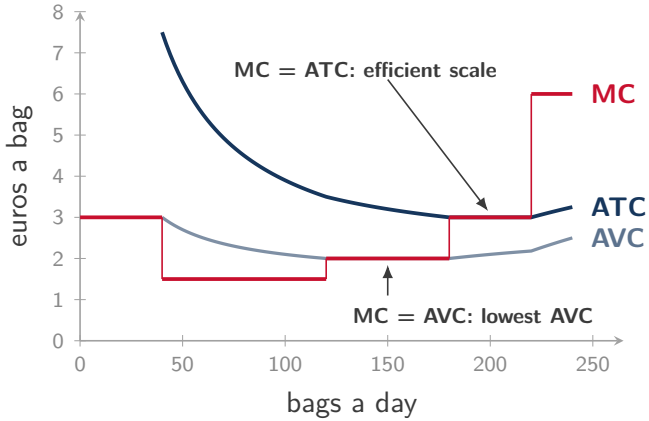


ATC = AFC + AVC: falling fixed cost per bag, then a crowded press.

Slide 15 Fixed cost spread thinner, a crowded press: average total cost is a U

Slide 10's table as curves, one at a time. The grey curve only ever falls, because the same rent is divided by more and more bags; the lighter blue one falls while the second printer's teamwork helps, runs flat, then rises as the press gets crowded. Average total cost is the two stacked, which is what the bracket shows at one output, so it is high at first because the rent sits on few bags and high again at the end because every bag needs more printer time; the U is the two forces taking turns. Our shop's bottom is flat because printers come in whole days; with half-days it would be rounder.

Marginal cost cuts through both averages at their lowest point



MC below the average pulls it down; above, pushes it up.

Slide 16 Marginal cost cuts through both averages at their lowest point

The last curve, and the rule that fixes where it crosses the others. Marginal cost falls while teamwork helps and then climbs steeply as the press gets crowded, the mirror of slide 8. Wherever the extra bag costs less than the current average, adding it drags the average down, and wherever it costs more, adding it lifts the average; so the average can only stop falling where the extra bag costs exactly the average, and that is where the two curves meet, at the bottom of each U. Think of a grade average and a new mark. The crossing with average total cost marks the efficient scale from slide 10; the same logic puts the other crossing at the bottom of average variable cost.

In pairs, two minutes: averages and the next one

Below the average or above it?

1. Average grade 6.0 after four exams; the fifth: 8.0. New average?
   6.4: above the average, it rises
2. A firm's ATC is 6.00 at 100 units; the next 20 cost 3.00 each. ATC at 120?
   5.50: below the average, it falls
3. Where MC equals ATC, ATC is...
   at its bottom: neither pulled down nor up

Slide 17 In pairs, two minutes: averages and the next one

Three quick checks of slide 16's rule, two with numbers and one in words. For the grade, the average is total points over exams, so the new average needs the old total back, not the old average and the new mark averaged together. For the firm, the same move with euros: the old total cost comes from the old average times the old output, the extra units add their own cost, and the new average divides the new total by the new output, not by the old one, and does not average the old average with the extra units' cost. The third item asks what the crossing point means in words, and the answer is the bottom of slide 15's U.

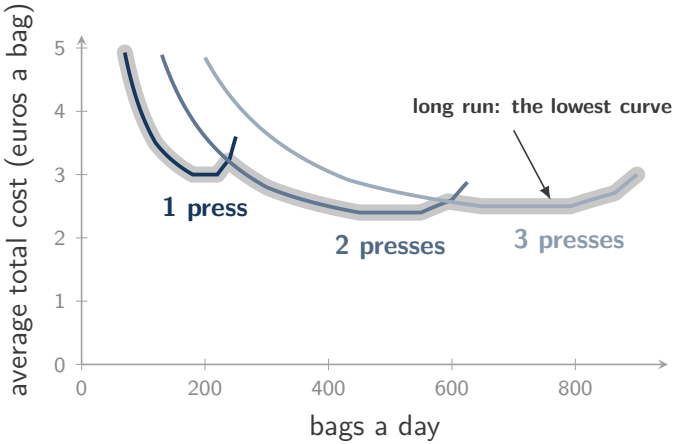
In the long run every cost can change: the shop can rent a second press

- Short run: the press is fixed; only printers change
- Long run: a second press, a bigger workshop
- In the long run every cost is variable
- How long? A print shop: months; a car plant: years

Slide 18 In the long run every cost can change: the shop can rent a second press

A definition slide, in words. The short run is however long some input stays fixed, here the press and the lease; the long run is long enough to change every input, so a cost that is fixed this year, the rent, becomes a choice in the long run. The split is not a number of months: it depends on how long the fixed input takes to change, which is why it is short for a print shop and years for a car plant. Everything before this slide was short run; the next three slides let the shop choose its size.

The long-run cost curve: the cheapest size for each output



In the long run the shop picks its size: the lowest curve at each output.

Slide 19 The long-run cost curve: the cheapest size for each output

Slide 15's U again, on a wider scale, next to the U the shop would have with two presses and with three. Each curve is a short run, a size the shop is stuck with; the long run is the choice among them, so for each output the shop would pick the size whose curve is lowest there, and the grey band traces those lowest pieces. Reading it: an order one press cannot fill at all becomes affordable with two, and a bigger size is only worth it once output is large enough to use it. With many sizes to choose from the scallops smooth into one flatter U, the reading's picture. The band falls, runs flat and rises again: the long run does not make cost fall for ever.

Double every input: if output more than doubles, the cost per bag falls

Now: 1 press, 4 printers: 220 bags for 660 a day, 3.00 a bag

Double every input: 2 presses, 8 printers, 1,320 a day.

if it makes	cost per bag	
550 bags	2.40	**economies of scale**
440 bags	3.00	**constant returns to scale**
330 bags	4.00	**diseconomies of scale**

What counts: does output grow faster than cost?

Slide 20 Double every input: if output more than doubles, the cost per bag falls

The long run in numbers, with the three names the reading gives. The shop doubles every input, so with the same input prices its cost doubles too; what decides the name is the race between output and cost, whether output grows faster, exactly as fast or slower, and the three rows show one case each. The cost per bag falls, stays or rises accordingly. The slip to avoid is reading economies of scale as total cost falling: total cost has doubled in every row; it is the cost per bag that moves. Slide 19's falling band was the first row; its rising end was the third.

Bigger is cheaper per bag — until coordination gets hard

Bigger, cheaper

workers who each do one task
bulk buying: ink by the barrel
machines that only come big
cheaper loans

Too big, dearer

more managers, slower decisions
messages lost on the way
workers far from the big picture

Slide 21 Bigger is cheaper per bag — until coordination gets hard

The reasons behind slide 20's names. Size lowers the cost per unit when it lets workers specialise, buys inputs in bulk, uses machines that only exist in big sizes and borrows more cheaply; the reading adds the economies a whole industry gets from clustering in one place. Size raises it when the firm becomes hard to run: more layers, slower decisions, messages lost between departments, workers who never see the finished product. Both cards hold for the same firm at different sizes, which is why the long-run curve of slide 19 first falls and later rises, and why a bigger firm is not always a cheaper one.

In pairs, five minutes: a laundry grows

A laundry for hotels

New machines take months to arrive.

	machines	staff	kilos a day	cost a day
now	4	2	300	600
doubled	8	4	750	1,200
doubled again	16	8	1,200	2,400

1. ATC now?
   2.00 euros a kilo
2. Doubled: ATC?
   What kind of returns?
   1.60: economies of scale
3. Doubled again: ATC? Returns?
   up to 2.00: diseconomies of scale
4. A hotel adds 100 kilos a day from next week. What can change by then?
   staff hours, not machines: the short run

Tote bags: which curve moves? (optional)

The market for printed tote bags

1. Printers' pay rises
   supply of totes shifts left
2. The charge for a plastic bag doubles
   demand for totes shifts right
3. Demand rises and totes get dearer: does supply shift too?
   no: they print more, along supply

Slide 22 In pairs, five minutes: a laundry grows

Exercise 3 is slides 18 to 20 on a laundry that doubles twice. The cost per kilo at each size is cost over kilos, and the kind of returns is read from the step between two sizes: inputs and cost double each time, so what matters is whether kilos grew by more or by less than that. The level the average ends at is not the test, only how it moved from the size before, which is why a number that returns to its starting value can still mean diseconomies. The last item is slide 18: within a week the hours of the staff can change but not the machines, so the extra kilos come with diminishing marginal product, whatever the laundry's plans to grow.

Slide 23 Tote bags: which curve moves? (optional)

An optional exercise on Class 5's shift question, tied to slide 9's input costs. A dearer input raises the cost of every bag printed, so the sellers' curve moves; a dearer substitute changes what buyers want at every price, so the buyers' curve moves; and the price of totes itself moves nobody's curve, only the point along it, which is the trap of the third item. The question to ask each time is which side's willingness at every price has changed, if either.

Cheap solar: positive or normative? (optional)

Cheap solar: three statements

1. "More solar panels bring more hours at a price of zero."
   positive: data can settle it
2. "Electricity should be cheap for everyone."
   normative: a judgement of values
3. A and B want cheap electricity. B: "Slow down: zero prices will scare off investors."
   B's reason: values or a prediction?
   a prediction: how investors respond

Slide 24 Cheap solar: positive or normative? (optional)

An optional exercise on Class 4's distinction, on the price from slides 5 and 14. A claim about what happens in the world is positive whenever evidence could in principle show it wrong, whether or not it turns out true; a claim about what ought to be weighs values and no data settles it. The third item is the trap: two people who share a goal can give opposite advice because they forecast differently how others will react, and a disagreement about a forecast is positive, not a clash of values.

A cost is what the firm gives up, paid out or not.

Once workers crowd the press, each adds less, so marginal cost rises; it cuts average cost at its lowest point.

In the long run every cost can change: economies of scale, then diseconomies.
Monday: how much a firm sells at a given price.

Slide 25 In one line

The class in four lines, to read slowly: the first is slides 6 and 7, the second slides 8 to 10 and 16, the third slides 18 to 21, and the fourth names Monday.

Before Monday 2 November

- Monday: perfect competition; monopoly begins
- Read Mankiw & Taylor: Ch. 5, pp. 123–137
- Problem set 2 (Block 3): your own version is on Campus Virtual; due Thursday 5 November, on paper
- Problem set 3 (Block 4): handed out Monday; due Thursday 12 November
- Practise in Sextant: the units the course page lists

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Slide 26 Before Monday 2 November

Monday 2 November: perfect competition, how much a firm like the print shop sells when it cannot set the price, and the start of monopoly. Reading: Mankiw & Taylor chapter 5, pages 123 to 137. Problem set 2, on Block 3, is on Campus Virtual (your own version, in the assignment Problem Set 2) and is due on Thursday 5 November, on paper, in class; Problem set 3, on Block 4, is handed out on Monday and due on Thursday 12 November. Practise in Sextant with the units the course page lists.

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